

12.328

EE25BTECH11002 - Achat Parth Kalpesh

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Question

For the given matrix

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix},$$

which of the following statements is/are true?

- ① $\mathbf{Q}$ is an orthogonal matrix
- ② $\mathbf{Q}^T = \mathbf{Q}^{-1}$
- ③ $\mathbf{Q}$ is a singular matrix
- ④ $\mathbf{Q}$ is a symmetric matrix

Theoretical Solution

$$\mathbf{Q}^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \quad (1)$$

$$\mathbf{Q}^T \mathbf{Q} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \quad (2)$$

$$\mathbf{Q}^T \mathbf{Q} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I} \quad (3)$$

Theoretical Solution

$$\det(\mathbf{Q}) = 1 \neq 0 \quad (4)$$

Also,

$$\mathbf{Q}^T \neq \mathbf{Q} \quad (5)$$

Thereby, $\mathbf{Q}$ is an orthogonal matrix i.e. $\mathbf{Q}^T = \mathbf{Q}^{-1}$